



لجنة الهندسة الميكانيكية
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أسئلة سنوات

كيمياء عامة (1) - ك101

First | Second | Final

2012

أسئلة سنوات



كتب



مانيوالات



Jordan University of Science & Technology
Department of Chemistry
CH 101 Second Exam
3/12/2012

الرقم المتسلسل:

الرقم الجامعي:

الإسم:

الشعبية:

المدرس: د.

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
A.	B	B	D	P	D	P	A	B	P	P	A	A	P	D	A	B	A	A	D	B	B

Avogadro's number = 6.03×10^{23}

- [Ar]4s²3d¹⁰4p¹ is the electron configuration of a(n) _____ atom.
A) Al B) Ga C) Ge D) In
- The electron configuration of a ground-state Ag atom is _____.
A) [Ar]4s²4d⁹ B) [Kr]5s¹4d¹⁰ C) [Kr]5s²3d⁹ D) [Ar]4s¹4d¹⁰
- The elements in the _____ period of the periodic table have a core-electron configuration that is the same as the electron configuration of krypton (Kr).
A) fifth B) second C) third D) fourth
- An electron cannot have the quantum numbers n = _____, l = _____, m_l = _____.
A) 2, 0, 0 B) 2, 1, -1 C) 3, 1, -1 D) 1, 1, 1
- The oxide of element will produce an acidic solution when dissolved in water.
A) Mg B) K C) Na D) N
- In which orbital does an electron in a phosphorus atom experience the greatest effective nuclear charge?
A) 1s B) 2s C) 2p D) 3s

7. Which of the following has the largest second ionization energy?
A) Na B) Mg C) Al D) P
8. The electron-domain geometry and molecular geometry of iodine trichloride (ICl₃) are _____ and _____, respectively.
A) trigonal bipyramidal, trigonal planar B) tetrahedral, trigonal pyramidal
C) trigonal bipyramidal, T-shaped D) octahedral, trigonal planar
9. The O-S-O bond angle in SO₂ is slightly less than _____.
A) 90° B) 109.5° C) 120° D) 180°
10. The hybridization of the central atom in the XeF₄ molecule is _____.
A) sp³d² B) sp² C) sp³ D) sp³d
11. The molecular geometry of the BrO₃⁻ ion is _____.
A) trigonal pyramidal B) trigonal planar C) bent D) tetrahedral
12. Of the molecules below, only _____ is polar.
A) CCl₄ B) CH₄ C) SeF₄ D) SiCl₄
13. The molecular geometry of the CHF₃ molecule is _____, and the molecule is _____.
A) seesaw, polar B) tetrahedral, nonpolar C) seesaw, nonpolar D) tetrahedral, polar
14. 24. Which of the following molecules or ions will exhibit delocalized bonding?
NO₂⁻ NH₄⁺ N₃⁻
A) NH₄⁺ and N₃⁻ B) NO₂⁻ only C) NO₂⁻, NH₄⁺, and N₃⁻ D) N₃⁻ only
15. The carbon-carbon σ bond in acetylene, HC≡CH, results from the overlap of _____.
A) sp hybrid orbitals B) sp³ hybrid orbitals C) sp² hybrid orbitals D) p atomic orbitals
16. In comparing the same two atoms bonded together, the _____ the bond energy the, _____ the bond order, and the _____ the bond length.
A) greater, smaller, shorter B) greater, greater, shorter
C) greater, greater, longer D) smaller, greater, greater
17. Based on molecular orbital theory, the bond orders of the F-F bonds in F₂⁺, F₂, and F₂⁻ are _____, respectively.
A) 2.5, 1, and 1.5 B) 1.5, 1, and 0.5 C) 1, 1.5, and 0.5 D) 0.5, 1.5, and 1
18. Which of these compounds is most likely to be ionic?
A) NF₃ B) CCl₄ C) SF₄ D) Mg₃N₂
19. Which of these ionic solids would have the largest lattice energy?
A) NaCl B) NaF C) CaBr₂ D) CaCl₂
20. Which of these atoms is the *least* electronegative?
A) Li B) Cs C) P D) As
21. Use bond dissociation energies to estimate the enthalpy change for the reaction of one mole of CH₄ with chlorine gas to give CH₂Cl₂ and hydrogen gas. CH₄ + Cl₂ → CH₂Cl₂ + H₂
D(C-H) = 413 kJ/mol; D(C-Cl) = 328 kJ/mol; D(H-H) = 436 kJ/mol; D(Cl-Cl) = 242 kJ/mol
A) -101 kJ/mol B) -234 kJ/mol C) +234 kJ/mol D) +106 kJ/mol

GOOD LUCK